

Divisibility

Coprime Numbers :

Pairs of numbers that have only one common factor 1

1) DIVISIBILITY BY 3:

M-1 If sum of all digits of a number is divisible by 3 then the whole number is divisible by 3

e.g. $783 = 7 + 8 + 3 = 18 \div 3 = 0$

$\therefore 783$ is divisible by 3

e.g. 7632189

$= 36 \div 3 = 0$ ✓ div. by 3

M-2 # for large nos
(by sum or observation)
search for 3/6/9 and cancel them and
do sum of digits and check divisibility
with 3

~~7~~~~6~~~~3~~~~2~~~~1~~~~8~~~~9~~ = $7 + 8 = 15 = 1 + 5 = 6 \div 3 = 0$

2) DIVISIBILITY BY 4:

$2^1 \rightarrow$ last 1 digit divisible by 2

$2^2 \rightarrow$ last 2 digits divisible by 4

i) 7284 62

$\rightarrow 7284 \div 4 \neq 0$

(X) not div.
by 4

ii) 5442829820 ✓

iii) 72568916400 ✓

3) DIVISIBILITY BY 6:

check divisibility by both 2 & 3

~~a) 12142722~~ $57.3 \div 3 \neq 0$

~~b) 5442829820~~

~~c) 792642~~

~~d) 92568916400~~

$792642 \div 2 = 0$

$7 + 9 + 2 + 6 + 4 + 2 = 31 \div 3$

$= 0$

4) DIVISIBILITY BY 8:

Last 3 digits div. by 8 or not

5) DIVISIBILITY BY 9:

Search for 9 in the no. & cancel them & check sum of remaining digits divisible by 9 or not

a) ~~2~~~~3~~~~5~~~~7~~~~7~~~~2~~~~9~~~~1~~~~3~~~~7~~ ✓

~~5~~~~2~~~~3~~~~5~~~~7~~~~2~~~~9~~~~1~~~~2~~~~4~~

~~4~~~~2~~~~5~~~~6~~~~8~~~~9~~~~1~~~~6~~~~4~~~~0~~~~6~~

$$8 + 6 + 4 + 0 + 6 \\ = 1800 \% 9 = 0$$

6) DIVISIBILITY BY 11:

~~8265~~

o Calculate sum of alternate digits

o Calculate their difference

o If difference $\div 11 = 0$ then no. is divisible by 11

M-①

$$8 + 6 = 14, 2 + 5 = 7$$

$$14 - 7 = 7 \neq 0$$

not divisible

~~b)~~ 216271

$$15 - 4 = 11 \neq 0 \checkmark$$

~~c)~~ 144282919

$$31 - 9 = 22 \neq 0$$

~~d)~~ 623521

$$11 - 8 = 3 \neq 0$$

~~e)~~ 52360

$$8 - 8 = 0$$

~~f)~~ 425686

$$17 - 0 = 17 \neq 0$$

M-2 : a)
$$\begin{array}{r} 52360 \\ +1+5 \\ \hline \end{array}$$

$$= 5 + 1 + 5$$

$$= 11 \div 11 = 0$$

$$\begin{array}{r} 11 \overline{) 66} \\ \underline{55} \\ 11 \\ \hline \end{array}$$

$$\begin{array}{r} 11 \overline{) 23} \\ \underline{22} \\ 1 \\ \hline \end{array}$$

Arrows point from the circled 5 in the first division and the circled 1 in the second division to the +5 and +1 in the first calculation.

b)
$$\begin{array}{r} 144282919 \\ \hline \end{array}$$

$$\begin{array}{r} 11 \overline{) 19} \\ \underline{22} \\ -3 \\ \hline \end{array}$$

$$+1-5-4-3 = -11 \div 11 = 0$$

An arrow points from the circled -3 in the second division to the -3 in the third calculation.

- o Always take that multiplier of 11 with which you will get smaller remainder
- o Then sum the remainders & check if divisible by 11 or not

7) DIVISIBILITY BY 12:

check divisibility by 3 & 4

a) ~~3~~1~~6~~272

$$\left. \begin{array}{l} \rightarrow 1 \div 4 = 0 \checkmark \\ \rightarrow 12 \div 4 = 0 \checkmark \end{array} \right\} \text{div. by } 12$$

8) DIVISIBILITY BY 15:

check divisibility by 3 & 5

9) DIVISIBILITY BY 7, 11, 13:

a) $\underline{005901}$

$$\begin{array}{r} 901 \\ - 5 \\ \hline 896 \end{array}$$

$\rightarrow$ div. by 7

$\rightarrow$ X div. by 11

$\rightarrow$ X div. by 13

- Make group of 3
- Find difference
- Check divisibility with difference

b) 045836

$$\begin{array}{r} 836 \\ 45 \\ \hline 791 \end{array}$$

✓ div. by 7

x div. by 11

x div. by 3

c) 004 572 568

+

$$568 + 4 = 572$$

$$\begin{array}{r} 572 \\ - 572 \\ \hline 0 \end{array}$$

0 → div. by 7
 → div. by 11
 → div. by 13

10) DIVISIBILITY BY 5

5 → last 1 digit divisibly by 5

25 → last 2 digit divisible by 5

ii) DIVISIBILITY By 101, 10101, 1001:

9

abab $\rightarrow$ divisible by 101

fermat

ababab $\rightarrow$ div. by 10101

abababab $\rightarrow$ div. by 1010101

abcabc $\rightarrow$ div. by 1001

1) 750750 $\rightarrow$ div. by 1001

$\rightarrow 750 - 750 = 0 \checkmark$

div. by 7, 11, 13